

**Operating Systems - CS(D)**  
**Quiz No. 4**

**Answer the following questions:**

Q1: What is a critical section and how do threads synchronize access to it?

Q2: What happens if you try to join a detached thread?

Q3: A system needs to check for updates from an external API every 10 minutes while the main application runs without interruption. Implement a detachable thread to perform this periodic check.

Q4: Modify Peterson's algorithm to track critical section entry counts for both threads

**Marks Distribution:**

$2 + 2 + 4 + 2 = 10$  Marks